NEOGUARD | ESP32-S3 SYSTEM CONNECTIONS

DRAFT E: direct signal wiring and ground symbols. USB-C battery power design is pending the battery specification.

MAX30102 / 0x57

J2

VCC 1 — V3V3
GND 2
SDA 3 — I2C_SDA_GPIO4 — 4 GPIO4_/_SDA
SCL 4 — I2C_SCL_GPIO5 — 5 GPIO5_/_SCL
Pin_5 5 — X
INT / unused

MLX90614 / 0x5A

J3

VCC 1 — V3V3
GND 2
SDA 3
SCL 4

DS1302 RTC

J4

VCC 1 — V3V3
GND 2
CLK 3 — RTC_CLK_GPIO6 — 6 GPIO6_/_CLK
DAT 4 — RTC_DAT_GPIO7 — 7 GPIO7_/_DAT
CE_/_RST 5 — RTC_CE_GPIO14 — 14 GPIO14_/_CE

EXTERNAL RGB LED

J7

RED_A 1 — LED_R_A — R3 330 — LED_R_GPIO16 — 16 GPIO16_/_RED
GREEN_A 2 — LED_G_A — R4 330 — LED_G_GPIO17 — 17 GPIO17_/_GREEN
BLUE_A 3 — LED_B_A — R5 330 — LED_B_GPIO18 — 18 GPIO18_/_BLUE
COMMON_K 4

330 ohms per colour; common cathode to GND.

U1

DEV3V3 → 3V3 → 3V3_OUT

ESP32-S3

DEVELOPMENT BOARD

GPIO00 0
GPIO01 1
GPIO02 2
GPIO03 3
GPIO013 13
GPIO019 19
GPIO020 20
GPIO035 35
GPIO036 36
GPIO037 37
GPIO038 38
GPIO039 39
GPIO043 43
GPIO044 44
GPIO045 45
GPIO046 46
GPIO047 47
GPIO048 48
EN EN
5V 5V

GPIO08_/_RST 8 — TFT_RST_GPIO8
GPIO09_/_DC 9 — TFT_DC_GPIO9
GPIO010_/_CS 10 — TFT_CS_GPIO10
GPIO011_/_MOSI 11 — TFT_MOSI_GPIO11
GPIO012_/_SCK 12 — TFT_SCK_GPIO12

GPIO021_/_CS 21 — SD_CS_GPIO21
GPIO040_/_SCK 40 — SD_SCK_GPIO40
GPIO041_/_MOSI 41 — SD_MOSI_GPIO41
GPIO042_/_MISO 42 — SD_MISO_GPIO42

GPIO15_/_BUZZ 15 — BUZZ_GPIO15

Logical GPIO interface
External connections
X = unused carrier pin

GND

ST7789 TFT / 240 x 280

J5

RST 5 — TFT_RST_GPIO8
DC 6 — TFT_DC_GPIO9
CS 7 — TFT_CS_GPIO10
MOSI 4 — TFT_MOSI_GPIO11
SCK 3 — TFT_SCK_GPIO12
GND 1
V3V3 2 — V3V3
BL_/_BLK 8 — TFT_BL

SPI MICROSD MODULE

J6

CS 6 — SD_CS_GPIO21
SCK 3 — SD_SCK_GPIO40
MOSI 4 — SD_MOSI_GPIO41
MISO 5 — SD_MISO_GPIO42
SD_VCC 1 — VCC
GND 2

BUZZER + DRIVER

BUZZ_BASE — R6 1k — BUZZ_GPIO15
R7 10k
Q1 (2)

D1 BUZZ_VCC 1 — BUZZ_SW 2

J8 POSITIVE / NEGATIVE

Q1 PN2222A: E1 / B2 / C3
D1 1N5819: stripe/cathode toward buzzer +
Active-high driver; buzzer 20 mA maximum.

Power and carrier harness

File: NeoGuard-power.kicad_sch

Selected V3V3 powers the modules; it is chosen on sheet 2. DEV3V3 is the controller output, not a short to external 3V3.

Default screen is the working ST7789 TFT. A different OLED needs its own verified interface. GPIO40-42 must be available on the actual board.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /
File: NeoGuard.kicad_sch

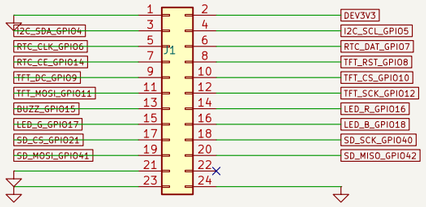
Title: NeoGuard – ESP32-S3 System Connections

Size: A3	Date: 2026-10-29	**Rev: E-DRAFT**
KiCad E.D.A. 10.0.4		Id: 1/2

NEOGUARD | POWER AND CARRIER HARNESS

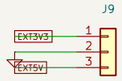
This sheet preserves the real carrier connector pin numbers and the selectable supply topology.

J1 / CONTROLLER CABLE



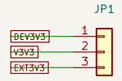
J1 is a cable connector, not a devboard socket.
Map each pin to the GPIO label on U1 / sheet 1.
Pin 2 is DEV3V3 from the ESP32 board.
Pins 1, 21, 23, 24 are ground; pin22 is unused.

INPUT SUPPLIES



J9 accepts already regulated supplies:
1 = external 3V3; 2 = GND; 3 = external 5V.
No onboard regulator or reverse-polarity protection.

JP1 / CARRIER 3V3 SOURCE



One shunt only:
1–2 = DEV3V3 -> selected V3V3
2–3 = EXT3V3 -> selected V3V3
Never fit both positions.

JP2 / SD MODULE SUPPLY

1–2 = V3V3; 2–3 = EXT5V.
One shunt only. Match the actual
SD breakout supply rating.

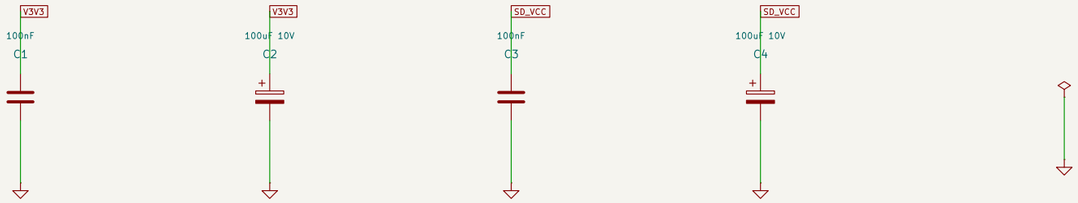
JP3 / BUZZER SUPPLY

1–2 = V3V3; 2–3 = EXT5V.
One shunt only. Match the buzzer
rating; this driver is for <=20 mA.

JP4 / TFT BACKLIGHT

Default OPEN / DNP.
Connect only for a verified 3V3-compatible
BL input with suitable driver/current limiting.

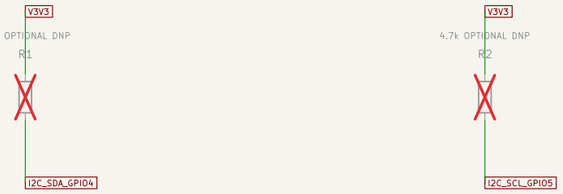
DECOUPLING / BULK STORAGE



Both copper layers use GND pours with thermals and stitching vias. Keep controller and modules powered together; signal isolation is not provided.

All carrier interfaces use 3.3V logic. Confirm breakout voltages, physical connector order, purchased-part pinouts and current budget before fabrication.

OPTIONAL I2C PULL-UPS



R1/R2 are DNP if module pull-ups already exist.
I2C pull-ups go to selected V3V3, never 5V.

GPIO names are logical signals; verify actual module headers and voltages

NeoGuard research prototype

Sheet: /Power and carrier harness/
File: NeoGuard-power.kicad_sch

Title: NeoGuard – Power and Carrier Harness

Size: A3	Date: 2026-10-29	**Rev: E-DRAFT**
KiCad E.D.A. 10.0.4		Id: 2/2